

① SELECT DISTINCT (department)
FROM students;

↓ return

department
IT
HR
Finance

② SELECT round(avg(age)) AS average-age,
department
FROM students
GROUP BY department;

↓ return

department	average-age
IT	21
HR	22
Finance	23

③ SELECT COUNT(DISTINCT student_id) AS student-count
department
FROM students
GROUP BY department
HAVING student-count > 1;

↓ return

student-count	department
2	IT
2	HR

④ SELECT student_id,
name,
age,
department

FROM students

WHERE age BETWEEN 21 AND 23;

↓ result

Student_id	Name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ SELECT *

FROM students

WHERE age > 21 AND (department = 'IT' OR department = 'HR');

↓

Student_id	name	age	department
2	Bob	22	HR
4	Diana	23	Finance
5	Eve	22	HR

⑥ SELECT SUM(credits) AS total_credits,
department

FROM ~~students~~ courses

GROUP BY department

HAVING total_credits > 5;

↓

total_credits	department
11	IT

⑦ SELECT *

FROM courses

WHERE ~~course~~ credits != 4;

⇓ return

course_id	course_name	department	credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧ SELECT course_id,
course_name,
credits

FROM courses

ORDER BY credits DESC

LIMIT 3;

⇓ return

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

⑨ SELECT MIN(grade) AS min-grade,
MAX(grade) AS max-grade,
AVG(grade) AS avg-grade
FROM enrollments;

⇓ returns

min-grade	max-grade	avg-grade
78	90	84.6(88)

⑩ SELECT
 COUNT(DISTINCT enrollment_id) AS enrollment_count,
 course_id
 FROM enrollments
 GROUP BY course_id;

⇓ return

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

⑪ SELECT department,
 SUM(salary) AS total_salary,
 SUM(bonus) AS total_bonus
 FROM salaries
 GROUP BY department;

⇓

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

⑫ SELECT department, AVG(salary) AS avg_salary
 FROM salaries
 GROUP BY department
 HAVING avg_salary > 55000;

⇓ return

department	avg_salary
IT	61 000
Finance	70 000

④ SELECT employee_id,
name,
salary,
bonus,
(salary + bonus) AS total-compensation

FROM salaries
WHERE (salary + bonus) > 60000;

⇓ return

employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

④ SELECT department,
sum(budget) AS total-budget,
AVG(budget) AS avg-budget

FROM projects

GROUP BY department

HAVING avg-budget > 70000;

⇓ return

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

⑤ SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';

⇓

Project-id	Project-name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
5	HR Portal	HR	50000